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CSE422 Case Study Assignment

2

Topic: Discriminatory Use of AI in Law Enforcement and Predictive Policing

Name: Tahsin Raihan Robbani
Id: 22201344
Section: 17

1. What does the application in the case study entail? Provide a detailed summary of the application, and provide references to papers with similar applications if you find any

Answer: The case study discusses the practical use of Artificial Intelligence (AI) in law enforcement, including predictive policing and risk assessment systems. Predictive policing uses algorithms to examine previous crime data and anticipate where future crimes are likely to occur or who may be involved in criminal activity. **According to Lum & Isaac (2016)**, this is accomplished by processing enormous volumes of data, such as geographic information, past crime reports, and social aspects such as demographics or socioeconomic status. Predictive tools aim to optimize the deployment of law enforcement resources by predicting crime patterns, reducing response times, and preventing crime in high-risk areas. The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) tool is a well-known example of this predictive policing in action. **According to ProPublica (2016),** COMPAS calculates the possibility of recidivism by producing a score based on factors such as age, family history, and criminal history. It has been widely used to decide sentencing and parole in the US criminal justice system. However, racial bias has been a point of criticism for COMPAS and other predictive tools. Black defendants were disproportionately identified as high-risk for reoffending by COMPAS, even when they did not commit additional offenses, whereas white defendants with similar criminal histories were often classified as lower risk **(ProPublica, 2016).** This demonstrates how algorithmic tools may unintentionally contribute to structural inequalities within the legal system. Numerous cities around the world have implemented comparable predictive policing applications. Programs such as PredPol, which forecast crime hotspots, make claims about reducing crime and improving law enforcement effectiveness **(Shroff, 2019**). But these systems can reinforce biases present in historical arrest data, giving minority or low-income communities. In the end, the use of these AI technologies requires careful oversight and ethical consideration because they have the potential to increase police effectiveness while also unintentionally reinforcing social bias.

2. What are the ethical implications of the application mentioned in the case study? Is there any other paper that talks about problems related to a similar domain?

Answer: The use of AI in law enforcement has significant ethical implications, especially when it comes to risk assessment and predictive policing. These tools usually raise significant questions regarding accountability, transparency, and fairness, even though their goal is to boost productivity and reduce crime.

- **Bias and Discrimination**: Most of the time, predictive policing algorithms learn from past data, which may have built-in biases. **O'Neil (2016)** asserts that elevated arrest rates and a precedent of excessive policing, particularly in minority communities, heighten the probability of an area being classified as high-risk. This strengthens racial disparities in the criminal justice system and encourages the cycle of excessive policing. **Angwin et al. (2016)** specifically demonstrated that COMPAS consistently overestimates the likelihood of re-offending among Black defendants while underestimating it for white defendants, highlighting how AI can exacerbate institutional racism.

- **Lack of Transparency:** Many AI-powered policing tools are private and closed-source. It is difficult for people to challenge potentially unfair or discriminatory outcomes because neither the general public nor law enforcement officials fully comprehend how these systems work **(Angwin et al., 2016)**. This lack of transparency raises ethical concerns about accountability in the legal system.
- **Privacy and Human Rights:** Gathering and evaluating vast amounts of data, including private and sensitive information, is necessary for predictive policing. According to **Eubanks (2018)**, widespread surveillance made possible by AI tools can have a disproportionately negative effect on people of color, which raises privacy and civil liberties concerns.

In summary, while AI technologies in law enforcement are predictive and effective, they also present serious ethical risks, such as bias, a lack of transparency, and privacy issues, which must be appropriately addressed to avoid systemic injustices.

3. **If such an application is deployed on a global scale, what could be the long-term future impact on society? Provide your opinion on this, and if possible, provide suitable references to support your argument**

Answer: There may be serious repercussions if predictive policing technologies like COMPAS and PredPol are implemented globally.

- **Increase of Social Inequalities: According to Brantingham & Brantingham (2008**), artificial intelligence (AI) systems that are based on historical crime data have the potential to reinforce existing inequalities. So, worldwide deployment may systemically harm underprivileged areas, resulting in protracted cycles of overpolicing and social disadvantage**.**
- **Erosion of Public Trust:** By undermining public trust in law enforcement, biased predictive policing can lead to social unrest and resistance, especially in communities that feel unfairly singled out (**Veale and Edwards, 2017)**. As a result, communities unfairly targeted may lose confidence in law enforcement, reducing cooperation and potentially triggering social unrest.
- **Privacy and Surveillance Concerns: According to Eubanks (2018),** the extensive application of predictive policing could lead to universal monitoring systems that continuously track people's movements and activities. This would raise serious concerns regarding civil liberties. So basically, it creates a “Big Brother” scenario, constantly monitoring personal movements and associations.
- **Legal and Policy Challenges: Edwards and Veale (2017)** says that putting these systems in place all over the world would lead to complicated arguments about fairness, oversight, and regulation. This is because some countries with weak oversight might abuse these systems, while others might enforce strict controls. This leads to differences in the results of justice.

In conclusion, AI can enhance law enforcement; however, if not carefully regulated, it may exacerbate inequality, threaten human rights, and undermine public trust in the rule of law.

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